

Arabic Maqam Identification via Generative Score-Matching and Regularized Joint Latent Representations

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Abstract

This report details a systematic investigation of several machine learning and algorithmic paradigms developed to solve the problem of Arabic Maqam identification. Utilizing a diverse dataset of instrumental solo performances and vocal compositions, we demonstrate the challenges of processing noisy audio representations under strict, non-overlapping dataset configurations to prevent data leakage. We evaluate traditional sequence modeling, classical heuristics, and generative diffusion-based score-matching metrics. Finally, we establish a highly robust joint latent optimization framework, combining a convolutional autoencoder and a dense classifier under a regularized loss formulation to reach an optimal identification accuracy of 93.2%.

I. INTRODUCTION AND PROBLEM FORMULATION

A. The Arabic Maqam System

In Arabic music theory, a *maqam* (plural *maqamat*) is the foundational system of melodic modes. Unlike Western equal-tempered scales, a maqam is characterized by non-tempered tuning, featuring microtonal intervals such as quarter-tones (neutral seconds). Furthermore, a maqam is not merely a static collection of pitches; it defines characteristic melodic trajectories, regional intonations, ornamentation patterns, and established rules of modulation (*sayr*).

Because many maqamat share identical scale intervals while differing solely in their melodic progression, classical scale analysis is insufficient. To make the problem tractable, we focus on classifying songs into broader *Maqam families* (e.g., the Rast, Bayati, or Sikah families), thereby narrowing down the classification space to distinct modal groupings.

B. Dataset Composition

Our dataset is constructed to reflect the stylistic and acoustic diversity of Arabic musical traditions. It consists of:

- **Oud Tanghim Solos:** Highly expressive, monophonic, and heterophonic instrumental improvisations with dynamic microtonal inflections.
- **Vocal Compositions:** Complex vocal recordings ranging from traditional choral ensembles to operatic performances, introducing significant vocal vibrato and diverse acoustic backdrops.

This combination ensures that our models must generalize across diverse instrumental timbres, noisy environments, and varied vocal ranges.

C. Data Preprocessing and Leakage Prevention

The raw audio signals are transformed into Mel spectrograms to represent the time-frequency energy of the recordings. To construct our training and evaluation samples, the long spectrograms are sliced into short temporal segments of length Δt .

To ensure complete statistical independence between samples and prevent severe data leakage, we enforce a strict **non-overlapping chunking scheme**. If chunks from the same recording were allowed to overlap, neighboring slices would share highly correlated temporal structures, artificially inflating validation accuracy during training. By segmenting the audio files with zero overlap and ensuring that all chunks from any single recording remain strictly within the same data split (train, validation, or test), we maintain a rigorous evaluation setup.

II. PRIOR APPROACHES AND HEURISTICS

A. Black-Box Sequence Modeling (LSTM)

Our first attempt to solve this problem utilized a sequence-based Long Short-Term Memory (LSTM) network. LSTMs are a common choice for music sequence classification and have shown success in identifying Turkish makams [4]. We implemented this approach on our dataset, extracting temporal pitch-class profiles to feed directly into the recurrent model.

However, testing this method on our noisy Arabic dataset exposed structural vulnerabilities. When evaluated by another member of our laboratory, the LSTM model struggled to generalize, achieving a classification accuracy of approximately 65% at best. While some runs approached 70% accuracy, this was found to be caused by subtle data leakage across overlapping test samples. In a mathematically rigorous, leak-free setup, the LSTM structure is highly sensitive to absolute pitch transpositions and background acoustic noise.

B. Algorithmic Heuristics

Prior to utilizing heavy machine learning models, we evaluated a rule-based, algorithmic heuristic to identify the active maqam. This approach relies on extracting fundamental frequency (f_0) trajectories and checking them against analytical scale definitions.

This non-parametric approach left us with an excessive number of arbitrary hyperparameters to tune, such as frequency tolerance windows, sliding window sizes, and voice-activity detection thresholds. Due to these hand-crafted dependencies, the algorithm was highly unstable, yielding volatile accuracies around $70\% \pm 5\%$ depending on the specific tuning of the heuristics.

III. DIFFUSION MODELS AS CLASSIFIERS

A. Score-Based Generative Frameworks and Parameterized Metric Optimization

A.1 Continuous-Time Stochastic Diffusion

To mitigate the high levels of noise present in vocal and live instrumental recordings, we model the spectrogram manifold through the framework of continuous-time generative diffusion processes [5]. Let $x_0 \in \mathbb{R}^D$ be a clean Mel spectrogram vector drawn from the empirical data distribution $p_{\text{data}}(x)$. We define the forward perturbation process over a finite time horizon $t \in [0, T]$ as a stochastic process governed by the following linear stochastic differential equation (SDE) [3]:

$$dx_t = f(x_t, t)dt + g(t)dw_t$$

where $f(\cdot, t)$ is a drift coefficient, $g(t)$ is a diffusion coefficient, and $w_t \in \mathbb{R}^D$ is a standard Brownian motion. Under this perturbation curriculum, the transition distribution $q(x_t | x_0)$ remains tractable and is given by:

$$x_t = \phi(x_0, t) = \alpha_t x_0 + \sqrt{1 - \alpha_t^2} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

where $\alpha_t \in [0, 1]$ represents a deterministic variance schedule defining the noise trajectory.

A.2 Score Formulation and Conditional Recovery

The conditional score function of the perturbed data distribution at scale t is defined as the spatial gradient of the conditional log-density:

$$s_\theta(x_t, t, y) \approx \nabla_{x_t} \log p_t(x_t | y)$$

where $y \in \{1, \dots, K\}$ is the conditional Maqam class label vector. Estimating $p_t(x_t | y)$ analytically is mathematically intractable. However, standard denoising score-matching establishes that the conditional score function can be directly estimated via a trained parameterized noise predictor $\epsilon_\theta(x_t, t, y)$ through the relationship [3]:

$$s_\theta(x_t, t, y) = -\frac{\epsilon_\theta(x_t, t, y)}{\sqrt{1 - \alpha_t^2}}$$

This parameterization allows us to define a conditional recovery mapping ψ_θ that projects the noisy state $x_t = \phi(x, t)$ back to its recovered, denoised state $\tilde{x}_y$ under a specific class hypothesis y :

$$\tilde{x}_y = \psi_\theta(\phi(x, t), t, y) = \frac{1}{\alpha_t} \left(\phi(x, t) - \sqrt{1 - \alpha_t^2} \epsilon_\theta(\phi(x, t), t, y) \right)$$

This reverse trajectory establishes a bidirectional mapping—resembling a Schrödinger bridge—between the tractably sampled Gaussian noise distribution $\mathcal{N}(0, \mathbf{I})$ and our empirical spectrogram distribution.

A.3 Parameterized Metric Space Mapping

For an unknown spectrogram x , the class-inference problem is formulated as finding the class label y that minimizes the reconstruction discrepancy between the input x and its conditionally recovered counterpart $\tilde{x}_y$. We evaluate this reconstruction error under a parameterized inner product space governed by a learnable metric matrix $A \in \mathbb{R}^{D \times D}$:

$$\langle x, \tilde{x}_y \rangle_A = x^T A \tilde{x}_y$$

Rather than utilizing heuristic thresholding boundaries, we project our metric comparisons into a valid probability distribution using the softmax operator σ . Thus, the comparative task of finding the class y that minimizes this distance is mapped directly to a parametric optimization problem over the parameters A :

$$\min_A \mathcal{L}(A) = -\frac{1}{N} \sum_{i=1}^N y_i^T \log(\sigma(x_i^T A \tilde{x}_i))$$

where $y_i \in \{0, 1\}^K$ is the one-hot encoded ground-truth target label vector for sample i , the logarithm is applied element-wise, and σ denotes the softmax operator.

B. Score-Matching Classifier Metric and Parametric Optimization

To establish a mathematically rigorous optimization framework, we define the following vectors for each sample:

- $x \in \mathbb{R}^D$: The original input spectrogram vector.
- $\tilde{x} \in \mathbb{R}^D$: The conditional class recovery vector, defined as $\tilde{x} = \psi_\theta(\phi(x, t), t, y)$.
- $y \in \{0, 1\}^K$: The one-hot encoded ground-truth class label vector.
- $f = \sigma(x^T A \tilde{x}) \in \mathbb{R}^K$: The predicted probability vector computed by applying the softmax operator σ .

We define the classification loss over N samples using the categorical cross-entropy metric:

$$\mathcal{L}(A) = -\frac{1}{N} \sum_{i=1}^N y_i^T \log(\sigma(x_i^T A \tilde{x}_i))$$

To minimize this loss with respect to the metric parameters A , we analyze the partial derivative of the loss term $\partial \mathcal{L}$. Let $f = \sigma(x^T A \tilde{x})$. Using the chain rule, we express the partial derivative of the loss composed at the function f :

$$\partial \mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \partial(y_i^T \log(p(y_i | x_i, t)))$$

We express this partial derivative as composed at the softmax output $f = \sigma(x^T A \tilde{x})$:

$$\partial \mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[\partial (y_i^T \log(f)) \Big|_{f=\sigma(x_i^T A \tilde{x}_i)} \right] \partial (x_i^T A \tilde{x}_i)$$

Evaluating the derivative of the logarithm under the softmax transformation yields the difference term $(f - y)^T$. We substitute this composition directly into the expression:

$$\partial \mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[(f_i - y_i)^T \right] (\sigma(x_i^T A \tilde{x}_i)) x_i^T \partial A \tilde{x}_i$$

$$\partial \mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[(\sigma(x_i^T A \tilde{x}_i) - y_i)^T \right] x_i^T \partial A \tilde{x}_i$$

We now rewrite the matrix variation term $x_i^T \partial A \tilde{x}_i$ using the Kronecker product identity $(\tilde{x}_i^T \otimes x_i^T) \partial A$:

$$\begin{aligned} \partial \mathcal{L} &= -\frac{1}{N} \sum_{i=1}^N \left[(\sigma(x_i^T A \tilde{x}_i) - y_i)^T \right] (\tilde{x}_i^T \otimes x_i^T) \partial A \\ &= -\frac{1}{N} \sum_{i=1}^N \left[(\sigma(x_i^T A \tilde{x}_i) - y_i)^T \right] (\tilde{x}_i \otimes x_i)^T \partial A \\ &= -\frac{1}{N} \sum_{i=1}^N \left[(\tilde{x}_i \otimes x_i) (\sigma(x_i^T A \tilde{x}_i) - y_i) \right]^T \partial A \end{aligned}$$

Since the total derivative of the loss is defined as $\partial \mathcal{L} = (\nabla_A \mathcal{L})^T \partial A$, we can directly isolate the final derivative matrix ∂_A^T :

$$\nabla_A \mathcal{L}^T = -\frac{1}{N} \sum_{i=1}^N \left[(\tilde{x}_i \otimes x_i) (\sigma(x_i^T A \tilde{x}_i) - y_i) \right]$$

Using first-order gradient descent, the metric matrix A is updated at iteration step $n + 1$ using the learning rate parameter η :

$$A^{(n+1)} = A^{(n)} - \eta \nabla_A \mathcal{L}(A^{(n)})$$

Our implementation of this score-based metric classifier achieved an identification accuracy of 56.0%. Although optimizing the geometry of the metric space A could improve classification performance, the minimization process is highly computationally expensive, requiring backpropagation through the reverse diffusion process for every candidate class.

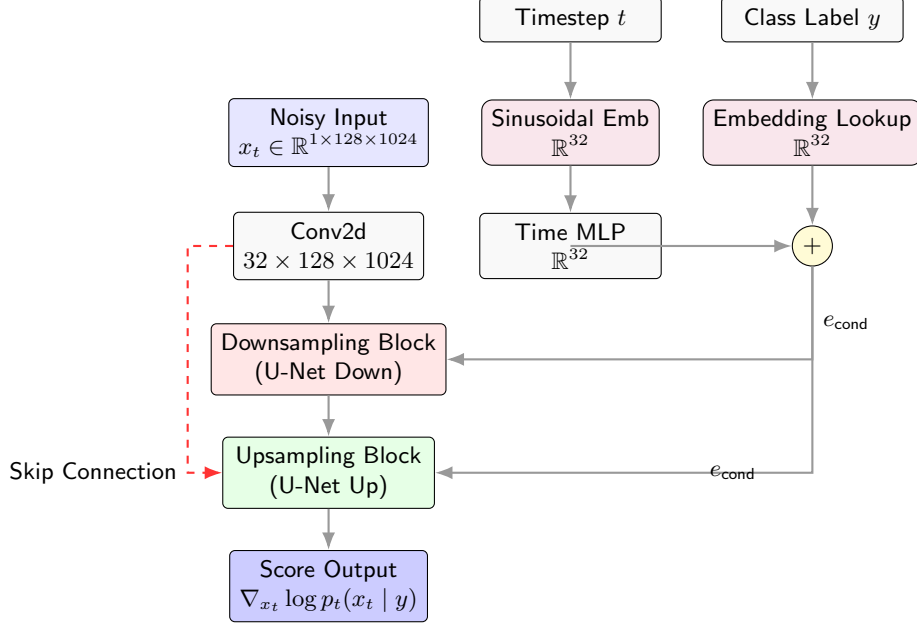


Figure 1: Architecture of the Conditional Denoising U-Net score predictor. Temporal embeddings and class representations are fused additively to form a conditioning vector, directing downstream convolutional blocks alongside spatial skip-connections.

IV. LATENT SPACE CLASSIFICATION

To achieve a more computationally efficient yet rigorous solution, we turned to latent space classification using a dual-network setup consisting of a convolutional autoencoder and a fully connected network (FCN) classifier. We evaluated three distinct loss formulations to study their impact on the learned representations.

A. Formulation A: Disjoint Reconstruction

We first trained a convolutional autoencoder to compress the input spectrogram $x \in \mathbb{R}^{B \times 1 \times 128 \times 1024}$ into a bottleneck latent representation $z \in \mathbb{R}^{B \times 128 \times 4 \times 32}$. The autoencoder was optimized strictly on reconstruction Mean Squared Error (MSE). Once trained, the encoder weights were frozen, and a dense classifier was trained on the static latent representations z . This decoupled setup yielded a limited accuracy of 66.0%, as pure MSE reconstruction forces the encoder to discard fine, low-amplitude microtonal lines.

B. Formulation B: End-to-End Shared Classification Loss

To prevent the encoder from losing class-relevant details, we detached the decoder and trained the encoder and the FCN classifier jointly using only the categorical cross-entropy loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(y, \hat{y})$$

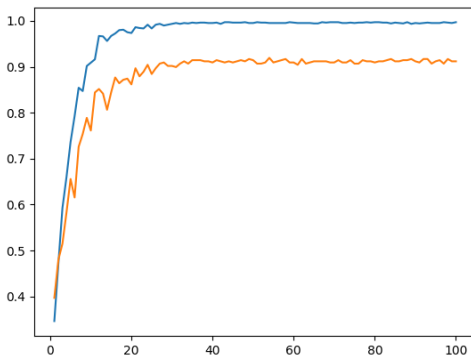
Under this setup, both networks share the same loss, forcing the convolutional layers of the encoder to learn features optimized solely for classification. This adjustment improved our identification performance to 84.0%.

C. Formulation C: Regularized Joint Loss

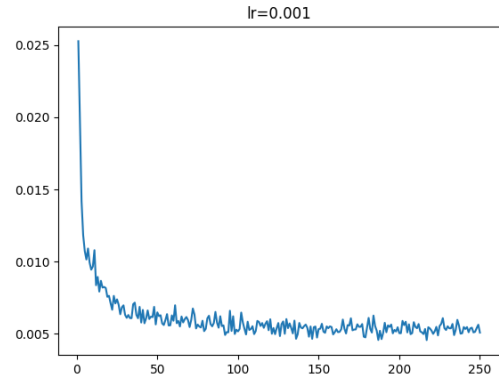
Finally, we implemented a joint multi-objective loss function. The classifier minimizes classification error, while the autoencoder minimizes a weighted combination of reconstruction error and classification cross-entropy:

$$\mathcal{L}_{\text{Autoencoder}} = \|x - \hat{x}\|_2^2 + \lambda \mathcal{L}_{\text{CE}}(y, \hat{y})$$

where $\hat{x}$ is the reconstructed spectrogram, and λ is a regularization hyperparameter. This joint formulation forces the encoder to preserve key physical acoustic details while structuring the latent space along clear modal boundaries, achieving our peak accuracy of 93.2%.



(a) Training and Validation Accuracy Curves



(b) Hyperparameter Tuning for λ

Figure 2: Performance metrics: (a) Comparative train/test accuracy curves over training epochs, and (b) analysis of the classification accuracy relative to variations in the regularization hyperparameter λ .

Table 1: Systematic Evaluation of Architectural Formulations

Model Topology	Optimization Objective	Accuracy
Sequential LSTM	Direct Pitch Sequence Classification	60.0%
Algorithmic Heuristics	Parametric f_0 Threshold Matching	$70.0\% \pm 5.0\%$
Conditional Diffusion	Metric Gradient Score-Matching	56.0%
Dual-Network (Formulation A)	Disjoint Unsupervised Reconstruction	66.0%
Dual-Network (Formulation B)	End-to-End Cross-Entropy Backpropagation	84.0%
Dual-Network (Formulation C)	Classification-Regularized Joint Loss	93.2%

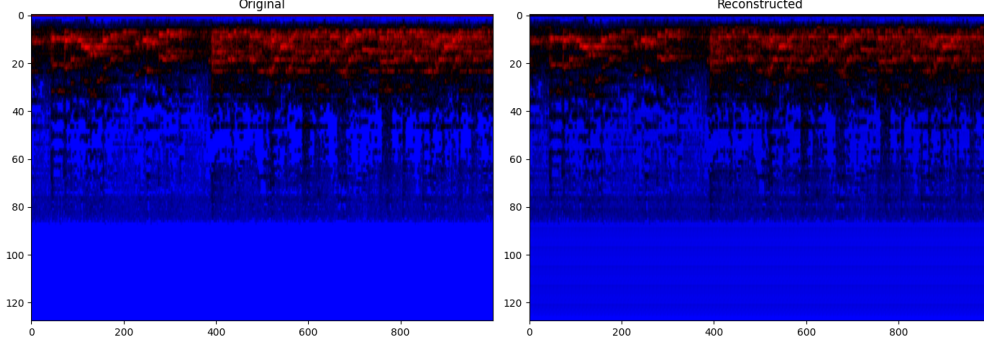


Figure 3: Original vs. reconstructed spectrogram under the joint loss framework. The regularized structure forces the encoder to preserve microtonal scale divisions alongside global structural paths.

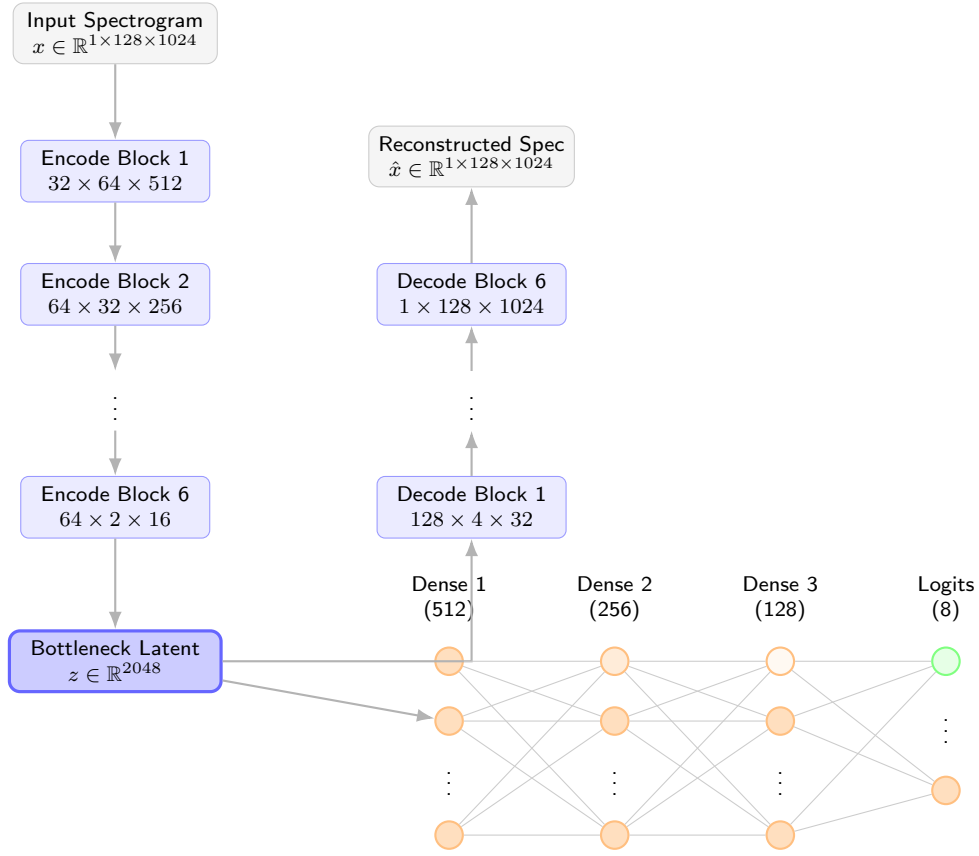


Figure 4: Architecture of the dual-network mapping. The input spectrogram is downsampled progressively through the convolutional encoder layers into a compressed bottleneck representation, which is mapped in parallel to the mirrored decoder (for spatial reconstruction) and the fully connected dense classifier (for maqam identification).

V. MECHANISTIC INTERPRETABILITY AND FUTURE WORK

Following the high accuracy achieved by our joint loss model, our current work focuses on analyzing how the network representations evolve during training. We are conducting mechanistic

interpretability studies to identify which specific neurons and convolutional kernels activate for different inputs.

By mapping these activation profiles back to acoustic frequency ranges, we aim to understand how the network isolates microtonal intervals and groups distinct melodies into their respective Maqam families.

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